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and progressive. Senescence is controlled by plant's own genetic programme. Death of the plant or its parts consequent to senescence is called **Programmed Cell Death** (PCD). The final stage of senescence is abscission. Abscission is a physiological process of shedding of organs from the parent plant body. The study of functioning of plants under adverse environmental conditions is called **stress physiology**. The environmental stress may broadly be divided into biotic and abiotic stress. The adverse effect on plants caused by other living organisms such as viruses, bacteria, fungi, parasites, insects, weeds is competing plants are called biotic stress. Abiotic stress may occur due to an atmospheric condition or soil condition.

Evaluation

1. Select the wrong statement from the following:
   a. Formative phase of the cells retain the capability of cell division.
   b. In elongation phase development of central vacuole takes place.
   c. In maturation phase thickening and differentiation takes place.
   d. In maturation phase, the cells grow further.
2. If the diameter of the pulley is 6 inches, length of pointer is 10 inches and distance travelled by pointer is 5 inches. Calculate the actual growth in length of plant.
   a. 3inches b. 6 inches
   c. 12 inches d. 30 inches
3. In unisexual plants, sex can be changed by the application of
   a. Ethanol b. Cytokinins
   c. ABA d. Auxin
4. Select the correctly matched one

A) Human urine	i) Auxin –B
B) Corn gram oil	ii) GA_3
C) Fungus	iii) Abscisic acid II
D) Herring fish sperm	iv) Kinitin
E) Unripe maize grains	v) Auxin A
F) Young cotton bolls	vi) Zeatin

   a) A-iii, B-iv, C-v, D-vi, E-i, F-ii,
   b) A-v, B-i, C-ii, D-iv, E-vi, F-iii,
   c) A-iii, B-v, C-vi, D-i, E-ii, F-iv,
   d) A-ii, B-iii, C-v, D-vi, E-iv, F-i
5. Seed dormancy allows the plants to
   a. overcome unfavourable climatic conditions
   b. develop healthy seeds
   c. reduce viability
   d. prevent deterioration of seeds
6. Which one of the following method are used to break the seed dormancy?
   a) Scarification b) Impaction
   c) Stratification d) All the above.
7. What are the parameters used to measure growth of plants?
8. What is plasticity?
9. Write the physiological effects of Cytokinins.
10. Describe the mechanism of photoperiodic induction of flowering.
11. Give a brief account on Programmed Cell Death (PCD)
12. What are the physiological effects of plants facing drought condition?
13. Explain the mechanism of biotic stress.